

Error	When It Happens	Example
SyntaxError	Invalid Python syntax	if x > 5 (<i>missing :</i>)
IndentationError	Incorrect indentation	if True:\nprint("Hi")
NameError	Using an undefined variable	print(age)
TypeError	Performing an operation on incompatible types	"10" + 5
ValueError	Correct type but invalid value	int("abc")
IndexError	List index out of range	numbers[5]
KeyError	Dictionary key doesn't exist	student["age"]
AttributeError	Object doesn't have that method/attribute	"hello".append("!")
ModuleNotFoundError	Module isn't installed or name is wrong	import numpyy
ImportError	Importing something that doesn't exist	from math import square
ZeroDivisionError	Dividing by zero	10 / 0
FileNotFoundError	File doesn't exist	open("data.txt")
PermissionError	No permission to access a file	open("/protected/file.txt")
RecursionError	Too many recursive function calls	A function repeatedly calling itself
AssertionError	An assert statement fails	assert age >= 18
OverflowError	Number is too large for an operation	math.exp(1000)
MemoryError	Program uses too much memory	Creating an extremely large list

Error	When It Happens	Example
EOFError	input() gets no input	Common in some online environments
KeyboardInterrupt	User stops the program	Pressing Ctrl + C
RuntimeError	A general runtime problem	Various runtime issues
NotImplementedError	A method is declared but not implemented	raise NotImplementedError()
StopIteration	Iterator has no more items	Calling next() after iteration ends
UnboundLocalError	Local variable used before assignment	Referencing a local variable before setting it
UnicodeEncodeError	Text can't be encoded	Saving unsupported characters
UnicodeDecodeError	Text can't be decoded	Reading a file with the wrong encoding
IsADirectoryError	Tried to open a directory as a file	open("folder")
NotADirectoryError	Expected a directory but got a file	Using a file path where a folder is expected
TimeoutError	Operation took too long	Network or socket timeout
ConnectionError	General network connection issue	Failed network connection
ConnectionRefusedError	Server refused the connection	Connecting to a closed port
BrokenPipeError	Writing to a closed connection	Sending data after the receiver closes
FloatingPointError	Floating-point calculation error (rare)	Certain numerical computations
BufferError	Buffer-related operation failed	Modifying an active buffer

Error	When It Happens	Example
SystemExit	Program exits using <code>sys.exit()</code>	<code>sys.exit()</code>
OSError	Operating system-related error	Disk, file, or OS operation failed